PUBLIC-SPECIFICATION HARDWARE PROJECT

MX tensor datapath

Five element formats. Exact arithmetic. Reproducible verification.

A parameterizable SystemVerilog block-dot processing element (PE) computes 32-element microscaled dot products. A broadcast array combines up to 16 PEs into a 4 x 4 output tile. This milestone implements and verifies RTL; SKY130 area, timing and power have **not** been measured.

Architecture and interface

Capability	Implemented contract
Element formats	FP8 E4M3/E5M2, FP6 E3M2/E2M3, FP4 E2M1
Scale and block size	Two E8M0 scales per block pair; 32 elements per block
Parallelism / depth	8, 16 or 32 lanes; 3, 4 or 5 pipeline stages
Arithmetic	Exact 70-bit signed accumulator or per-product TRUNC(n)
Outputs	FP32 or BF16, directly rounded ties-to-even; exception flags
Integration	Ready/valid streams, APB control, sticky flags, retired-tile counter

Four-stage baseline

Decode extracts finite significands and specials. **Multiply** uses at most 4 x 4-bit significand products and aligns them on a common binary grid. **Reduce** uses a recursive carry-save tree and accumulates earlier beats. **Convert** normalizes and rounds once to the selected destination.

A global pipeline enable freezes all stages when the output stalls. The no-bubble test demonstrates one completed block every 1, 2 or 4 cycles for 32, 16 or 8 lanes, respectively. These are cycle counts, not measured clock frequencies.

The key numerical decision

Exact summation avoids order-dependent intermediate rounding. Including the sign bit, the format-native 32-term accumulator widths are 42, 70, 24, 18 and 14 bits for E4M3, E5M2, E3M2, E2M3 and E2M1. Runtime format selection therefore retains a 70-bit datapath. Runtime truncation explores accuracy; it does not claim an area reduction.

TRUNC(n) removes low magnitude bits before applying the product sign, so negative values truncate toward zero. BF16 is rounded directly from the wide sum, avoiding FP32-to-BF16 double rounding. Subnormals are preserved. NaN scales ignore element encodings and poison the block.

VERIFICATION AND EVIDENCE

What actually passed

Check	Recorded result
Reference model	656 element encodings cross-checked against ml_dtypes; 44,000 rounding checks
PE regression	48,180 checked block outputs across 9 lane/depth configurations
Functional coverage	160/160 declared contract bins + 7/7 rounding classes in every configuration
Array and APB	90 tiles / 630 PE results across 1x1, 2x2 and 4x4 arrays
Mutation campaign	9/9 named RTL mutants killed; passing baseline required
Formal	70-bit CSA compressor identity; unbounded PE reset and stalled-output properties

How the evidence is produced

A pyuvm environment, agent and scoreboard use a cocotb BFM to reconstruct transactions from accepted DUT pins. An integer-only reference predicts data and flags. Coverage is awarded only after a matching result retires. Directed sequences close boundary bins; seeded random tests vary input values, scales, truncation, bubbles and backpressure.

The declared bins cross format, output, exact/TRUNC(1) mode and eight scenarios, plus seven rounding classes. Full coverage of this finite plan is **not** exhaustive arithmetic or code coverage. Formal results cover the named structural/protocol properties, not end-to-end floating-point equivalence.

A real bug found during development

A NaN scale produced the expected quiet NaN but also raised an approximation flag from discarded element bits. The scoreboard caught the flags mismatch. The RTL now suppresses element-derived approximation events when a scale is NaN. Planted defects separately exercise scale bias, subnormals, E4M3 specials, sign, ties, NaN scales, multi-beat summation, truncation and backpressure.

Reproduce and finish the physical milestone

After dependency setup: `make reproduce`. Raw logs, coverage, mutation results, tool versions and source hashes are in reports/. The OpenLane configuration and physical preflight are provided; a SKY130 PDK and complete OpenROAD/OpenSTA environment are still required. Routed parasitics and audited switching activity must accompany any future timing/power claim.

Public-specification scope

NVIDIA's internal tensor-core designs are proprietary. This project is built only from public specifications (the OCP Microscaling Formats spec and standard IEEE-754 behavior), so it shows the same class of problem without claiming to replicate any real product.

Reference: OCP Microscaling Formats v1.0 (September 2023), sections 5.2-5.4 and 6.1. Source link and exact contract: docs/microarchitecture.md. Physical and coverage limitations: physical/README.md and docs/verification-plan.md.